

# Lecture 14: Free energy

Jochen Braun

Otto-von-Guericke-Universität Magdeburg,  
Cognitive Biology Group

Theoretical Neuroscience II, SS 2026

Credits:

Karl Friston (2010) A unified brain theory? Nat. Rev. Neurosci.

Jakob Howhy (2017) How to entrain your evil demon. In: *Philosophy and Predictive Processing*.

Rafal Bogacz (2015) A tutorial on free-energy. J. Math. Psychol.

E.T. Jaynes (1967) Information theory and statistical mechanics.

## 15. Free energy

How can a brain 'understand' its environment? What would such an 'understanding' involve? Causal models offer plausible answers. External causes (true, physical) produce sensations. Brains try to explain sensations in terms of internal causes (hypothetical, mental). Hopefully, over time, internal causes come to correspond to external causes.

Many theories in neural and physical sciences can be viewed from a free-energy perspective (e.g., Hebbian learning, information theory, optimal control theory). In all cases the central theme is maximization of certainty (expected value, expected utility) combined with minimization of surprise (prediction error, expected cost). A free-energy principle is claimed to account for action, perception, learning, among others.

This lecture illustrates the free-energy principle in some concrete situations.

## 1. Plato's cave and causal models



1. Plato's cave and causal models
2. Causal models and free energy
3. A simple perceptual problem
4. Free energy in statistical physics
5. Derivation of Boltzmann distribution (E.T. Jaynes)

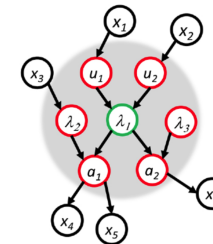
Plato's "Allegory of the Cave" illustrates the difference between direct and inferred knowledge. In the allegory, prisoners are chained together behind a wall in a cave. Behind the wall there is a fire, and between the fire and the prisoners there are people carrying puppets or other objects. These objects cast shadows on the cave wall, which the prisoners are able to see. As the prisoners see only these shadows, they believe them to be real. In fact, however, they see only shadows, never the objects directly.

In a similar way, we build perceptions only on sensations, never directly on objects themselves. However, unlike the unfortunate prisoners, we can test our perceptions by interacting with the perceived objects. Thus, we believe that our perceptions closely approximate the objects themselves.

Philosophers continue to debate the degree of "certainty" or "knowledge" that we can reach in this way.

## Markov blanket

Consider a system with (possibly circular) causal dependencies between nodes (arrows). A node  $\lambda_1$  is causally dependent on parent nodes ( $u_1, u_2$ ) and in turn causally influences children nodes ( $a_1, a_2$ ), which also depend on other nodes ( $\lambda_2, \lambda_3$ ).



Together these nodes form a "Markov blanket" for  $\lambda_1$ , as in principle its state is predictable just from observing blanket states, without knowing anything about external nodes.

## Perceptual and active inference

The brain models its Markov blanket and, indirectly, the environment. The model gathers evidence for itself by explaining away sensory input. In a process of approximate Bayesian inference, hypotheses about sensory input are generated from the model, and the predictions tested against the actual input.

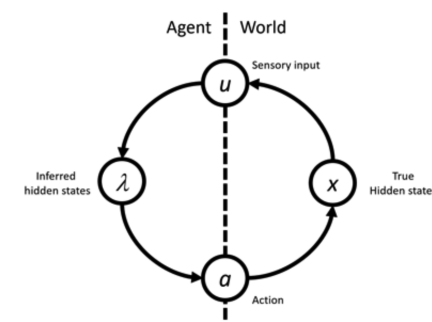
*Hypothesis: current sound is caused by approaching train.*

*Prediction: it will soon get louder and then recede.*

Sensory input can become more predictable by changing the active states of the Markov blanket. Certain active states produce certain sensory states.

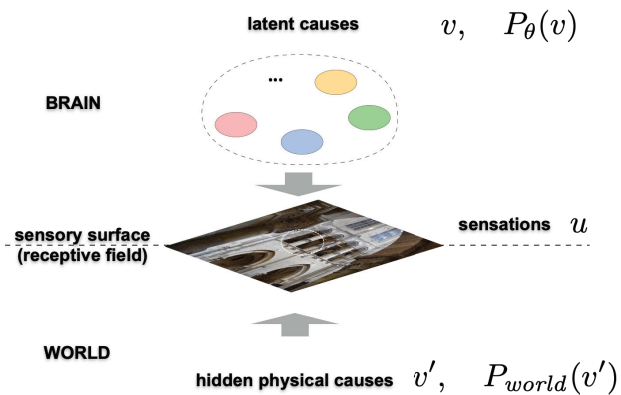
*Hypothesis: approaching train. Action: go to window and look.*

## Causal models of Markov blanket approximate true causes

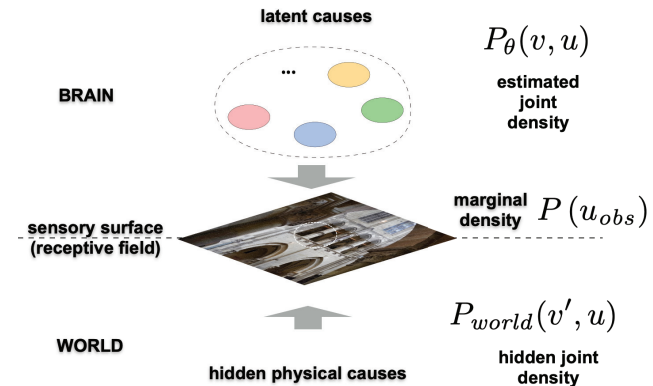


In perceptual inference, the internal model builds expectations for how sensory states unfold over time, for example, between sensory states  $u_1$  and  $u_2$ . In active inference, it will also predict associations between active state  $a_1$  and sensory state  $u_2$ , given internal states  $\lambda_1$ .

## 2a. Causal models and free energy (simplified)



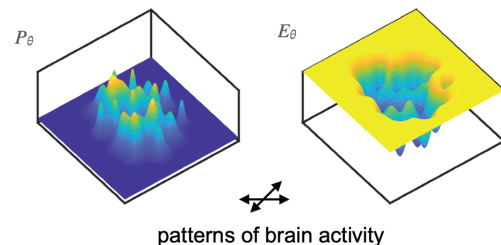
Brains implement causal models with 'latent causes'  $v$  and associated probabilities  $P_{\theta}(v)$ . Model parameters are in  $\theta$ .



The aim is to reproduce observed sensations,  $P(u_{obs})$ , in terms of marginal and joint distributions,  $P_{\theta}(v)$  and  $P_{\theta}(v, u)$ .

## Potential landscape for patterns of brain activity

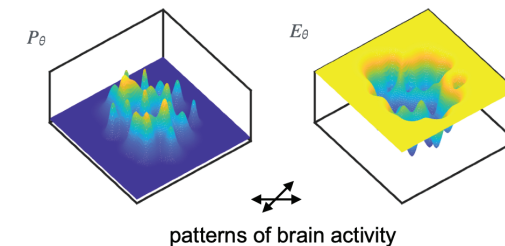
$$P_{\theta}(v, u) = Q^{-1} \exp[-E_{\theta}(v, u)], \quad Q = \sum_{v, u} \exp[-E_{\theta}(v, u)]$$



Assume that the dynamics of brain activity is governed by a potential landscape and by noise. If activity encodes latent causes  $v$  and predicted observations  $u$ , the probability of activity visiting particular combinations  $v, u$  is determined by this potential. This physical realization of probabilities is called "Langevin sampling".

## Equilibrium can be 'recognition distribution'

$$\frac{\exp[-E_{\theta}(v, u_{obs})]}{\sum_v \exp[-E_{\theta}(v, u_{obs})]} = \frac{P_{\theta}(v, u_{obs})}{\sum_v P_{\theta}(v, u_{obs})} = \frac{P_{\theta}(v, u_{obs})}{P_{\theta}(u_{obs})} = P_{\theta}(v|u_{obs})$$



When activity is constrained by observed  $u_{obs}$ , the equilibrium distribution becomes the 'recognition distribution' under the model,  $P_{\theta}(v|u_{obs})$ . Thus, the most likely activity states represent the most likely causes  $v$  of the given observations  $u_{obs}$ .

Dayan, Hinton, Neal, Zemel (1995) Helmholtz Machine. *Neural Comput.* 7(5): 889-904.

## Equilibrium distribution maximizes entropy

The entropy can be decomposed as follows

$$\underbrace{-\sum_v P_\theta(v|u_{obs}) \log P_\theta(v|u_{obs})}_{\text{entropy of explanation}} = -\sum_v P_\theta(v|u_{obs}) \log \frac{P_\theta(v, u_{obs})}{P_\theta(u_{obs})}$$

$$= \underbrace{\log P_\theta(u_{obs})}_{\text{likelihood of observation}} + \underbrace{\langle E_\theta(v, u_{obs}) \rangle}_{\text{expected energy}}$$

because  $\sum_v P_\theta(v|u_{obs}) = 1$

and  $-\log P_\theta(v, u_{obs}) = E_\theta(v, u_{obs}) + \log Q$

if we disregard the constant  $\log Q$ .

Dayan, Hinton, Neal, Zemel (1995) Helmholtz Machine. *Neural Comput.* **7**(5): 889-904.

## Equilibrium distribution minimizes ‘free energy’

Alternatively, we can use the identity  $1/P(u) = P(v|u)/P(v, u)$

$$-\sum_v P_\theta(v|u_{obs}) \log P_\theta(u_{obs}) = \sum_v P_\theta(v|u_{obs}) \log \frac{P_\theta(v|u_{obs})}{P_\theta(v, u_{obs})}$$

to express free energy (surprise) as follows

$$F_\theta \equiv \underbrace{-\langle \log P_\theta(u) \rangle_u}_{\text{surprise}} = \left\langle \sum_v P_\theta(v|u) \log \frac{P_\theta(v|u)}{P_\theta(v, u)} \right\rangle_{u_{obs}}$$

This expression provides a theoretical justification for “expectation maximization”, because we can minimize free energy by alternately improving a ‘recognition function’  $P_\theta(v|u)$  and a ‘generative model’  $P_\theta(u|v)$ , given a body of observations  $P(u_{obs})$ .

Dayan, Hinton, Neal, Zemel (1995) Helmholtz Machine. *Neural Comput.* **7**(5): 889-904.  
 Friston (2008) Hierarchical models in the brain. *PLoS Comput. Biol.*  
 Friston (2010) Free energy principle: a unifying brain theory? *Nat. Rev. Neurosci.*  
 Gottwald et al. (2020) Two kinds of free energy. *PLoS Comput. Biol.*

## Free energy for point estimate

When observation  $u$  leads to a point estimate for cause  $v = \phi(u)$ , the recognition density is a delta function  $P_\theta(v|u) = \delta[v - \phi(u)]$  and the free energy becomes

$$\begin{aligned} F_\theta &= \int dv P_\theta(v|u) \log \frac{P_\theta(v|u)}{P_\theta(v, u)} = \\ &= \int dv \delta[v - \phi(u)] \log \frac{\delta[v - \phi(u)]}{P_\theta(v, u)} = \\ &= -\log P_\theta[u, \phi(u)] + \underbrace{0}_{\text{entropy of } \delta[v - \phi(u)]} \\ &= -\log P_\theta[u|\phi(u)] + \log P_\theta[\phi(u)] \end{aligned}$$

This simplified expression will be used in Section 3.

## Weakness of ‘free energy’ framework

- ▶ Equilibrium framework assuming a ‘beginning’ and an ‘end’.
- ▶ Information is processed until an answer is found.
- ▶ Activity explores high-dimensional landscape and settles on region around minimum.
- ▶ Brain activity does not settle and is not at equilibrium (but continuously expresses unprompted and spontaneous changes).
- ▶ Perceptual states do not settle and are not at equilibrium (but continuously expresses unprompted and spontaneous changes).
- ▶ A non-equilibrium framework seems more plausible to me.

## “Langevin sampling” of posterior probability

Show movie.

## 3. A simple perceptual problem

We consider the problem of inferring a one-dimensional cause  $v$  from a one-dimensional observation  $u$ . Prior experience with such causes tells us that, firstly, they produce noisy sensory observations  $u$ , with distribution

$$p(u|v) = \mathcal{N}[u|\mu_u(v), \Sigma_u] = \frac{1}{\sqrt{2\pi\Sigma_u}} e^{-(u-\mu_u)^2/2\Sigma_u}$$

and, secondly, they have a prior distribution

$$p(v) = \mathcal{N}[v|\mu_v, \Sigma_v] = \frac{1}{\sqrt{2\pi\Sigma_v}} e^{-(v-\mu_v)^2/2\Sigma_v}$$

How should we interpret an observed  $u_{obs}$  in the light of this prior experience?

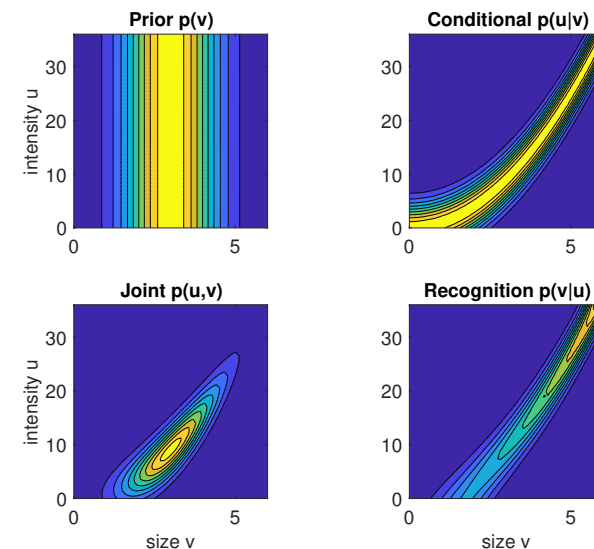
## Exact solution

We can use Bayes' theorem to compute the likelihood of different values  $v$ , given our observation  $u_{obs}$ :

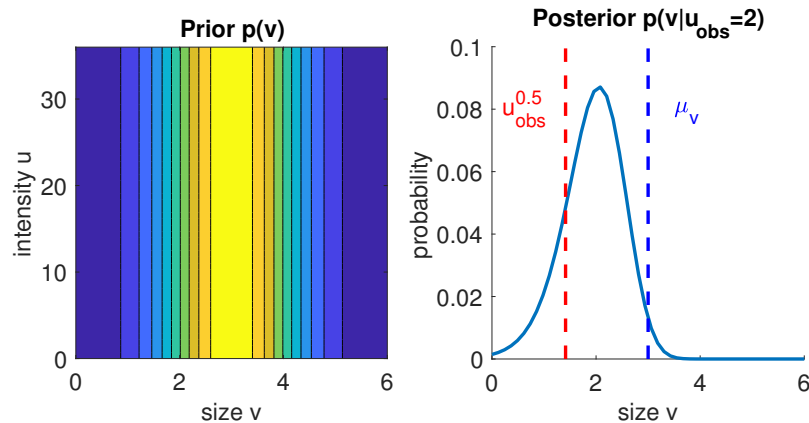
$$p(v|u_{obs}) = \frac{p(u_{obs}|v) p(v)}{p(u_{obs})}$$

Note that  $p(v)$  and  $p(u_{obs}|v)$  form part of our prior experience and that the marginal distribution  $p(u_{obs})$  is implied by this experience by integrating over all causes

$$p(u_{obs}) = \int p(u_{obs}|v) p(v) dv$$



Prior distribution  $p(v)$  with  $\mu_v = 3$ ,  $\Sigma_v = 1^2$ , conditional distribution  $p(u|v)$  with  $\mu_u = v^2$ ,  $\Sigma_u = 3^2$ . Joint distribution  $p(u, v)$  and recognition distribution  $p(v|u)$  follow.



The most likely value of  $v_{opt} \approx 2.2$  lies *between* the value suggested by the evidence,  $g^{-1}(u_{obs}) = \sqrt{u_{obs}} = \sqrt{2}$ , and the most probable *a priori* value,  $\mu_v = 3$ .

## Free energy

As the exact solution involves an integral, it is not easy to compute. Perhaps it can be approximated?

Change notation  $v_{est} \rightarrow \phi$  and seek to maximize  $p(\phi|u_{obs})$  or, equivalently,

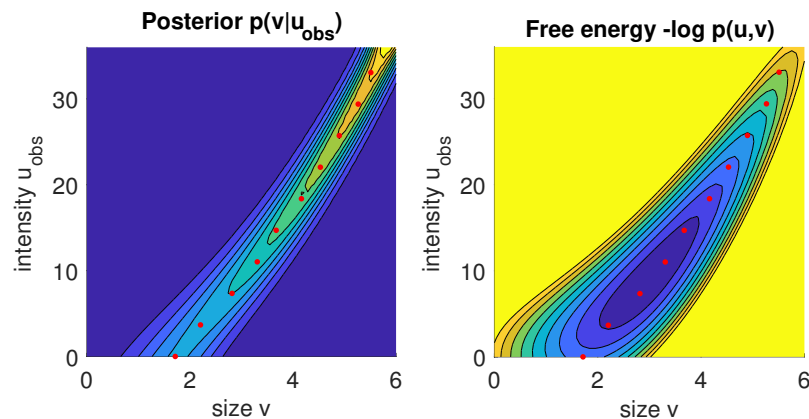
$$\ln p(\phi|u_{obs}) = \ln \frac{p(u_{obs}|\phi)p(\phi)}{p(u_{obs})} = \underbrace{\ln p(\phi) + \ln p(u_{obs}|\phi)}_{-F} - \ln p(u_{obs})$$

Note that only the  $\phi$ -dependent terms need to be considered, which correspond to negative 'free energy'

$$F = -\ln p(u_{obs}, \phi) = -\ln p(u_{obs}|\phi) - \ln p(\phi)$$

Thus, we may equivalently minimize free energy.

## Comparison of posterior and free energy



Maximizing the posterior probability is equivalent to minimizing free energy, because they differ by factor,  $p(u)$ , unchanged by  $v$ .

## Gradient descent

Assume that  $p(v)$  and  $p(u|v)$  are approximately Gaussian (and  $0 \leq p \leq 1$ ):

$$\begin{aligned} F &= -\ln p(u|v) - \ln p(v) = \\ &= -\frac{1}{2} \ln \Sigma_v - \frac{(\phi - \mu_v)^2}{2\Sigma_v} - \frac{1}{2} \ln \Sigma_u - \frac{[u - g(\phi)]^2}{2\Sigma_u} + 2 \ln \frac{1}{\sqrt{2\pi}} \end{aligned}$$

and consider how  $F$ , with  $-\infty < \ln p \leq 0$ , changes with value of posterior estimate  $\phi$ :

$$\frac{\partial F}{\partial \phi} = -\frac{\phi - \mu_v}{\Sigma_v} + \frac{u - g(\phi)}{\Sigma_u} g'(\phi)$$

$F$  grows with  $\phi$  as long as  $\phi$  is below some optimal value, and shrinks with  $\phi$  as soon as  $\phi$  exceeds this value.

## Dynamic implementation

To find the best  $\phi$ , we can start from some initial guess  $\phi_0$  and then change  $\phi(t)$  over time with a 'gradient ascent' rule:

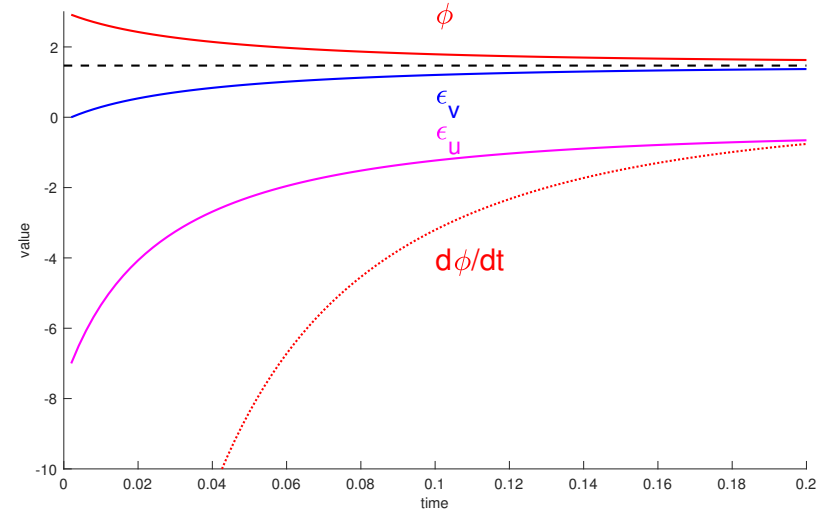
$$\dot{\phi} = \frac{d\phi}{dt} \approx \frac{\partial F}{\partial \phi}$$

This ensures that  $\phi$  grows (moves right) when this increases  $F$  (positive slope), but that  $\phi$  shrinks (moves left) when this increases  $F$  (negative slope).

Specifically, we implement the dynamical system

$$\frac{d\phi}{dt} = -\frac{\phi - \mu_v}{\Sigma_v} + \frac{u - g(\phi)}{\Sigma_u} g'(\phi)$$

and find that it converges from an initial guess  $\phi_0 = \mu_v$  to the theoretically correct value  $\phi_{opt} \approx 2.2$ .



## Prediction errors

The two conflicting constraints (on  $v$  and on  $u$ ) can be quantified in terms of distance from the mean, in units of variance. These can be considered 'prediction errors':

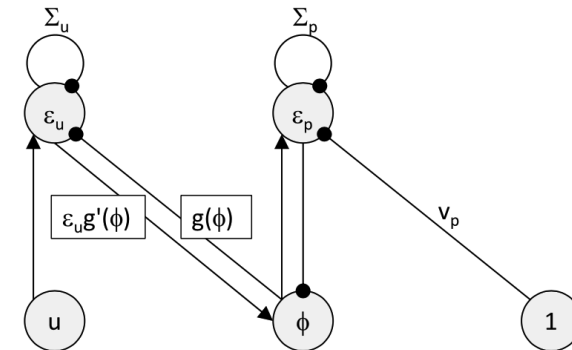
$$\epsilon_v = \frac{\phi - \mu_v}{\Sigma_v}, \quad \epsilon_u = \frac{u - g(\phi)}{\Sigma_u}$$

Extending our dynamical system, we can obtain also the prediction errors from dynamical interactions:

$$\dot{\epsilon}_v = -\Sigma_v \epsilon_v + \mu_v - \phi$$

$$\dot{\epsilon}_u = -\Sigma_u \epsilon_u + u - g(\phi)$$

The dynamic variables that depend on  $\phi$  are  $\phi$  itself,  $\epsilon_p$ , and  $\epsilon_u$ . The model parameters  $\mu_v$ ,  $\Sigma_p$ ,  $g(v)$ , and  $\Sigma_u$  are implemented as 'synaptic strengths'.



**Fig. 3.** The architecture of the model performing simple perceptual inference. Circles denote neural "nodes", arrows denote excitatory connections, while lines ended with circles denote inhibitory connections. Labels above the connections encode their strength, and lack of label indicates the strength of 1. Rectangles indicate the values that need to be transmitted via the connections they label.

Rafal Bogacz (2015) A tutorial on the free-energy framework in perception and learning. J. Math. Psychol.

## 4. Free energy in statistical physics

A macroscopic steady-state state  $X$  is described by a distribution  $p(x)$  over microscopic states  $x \in X$ , with  $\sum_{x \in X} p(x) = 1$ . Assume that each microscopic state  $x$  is associated with a potential  $\Phi(x)$ .

Constraining the average (expected) potential to some value  $\langle \Phi \rangle = \sum_x p(x) \Phi(x)$  while maximizing entropy  $S = -\sum_x p(x) \ln p(x)$  results in the Boltzmann distribution:

$$p(x) = \frac{1}{Z} e^{-\alpha \Phi(x)}, \quad Z \equiv \sum_x e^{-\alpha \Phi(x)}, \quad \langle \Phi \rangle = -\frac{\partial}{\partial \alpha} \ln Z$$

where  $\alpha$  is a negative temperature and  $Z$  is called the 'distribution function'. Note that  $p(x)$  decreases exponentially with the potential  $\Phi(x)$ .

## Entropy and free energy

For a given energy value  $\langle \Phi \rangle$ , the maximal entropy can be rewritten

$$S_{max} = -\sum_x \underbrace{\frac{1}{Z} e^{-\alpha \Phi(x)}}_{p(x)} \underbrace{[-\ln Z - \alpha \Phi(x)]}_{\ln p(x)} = \ln Z + \alpha \langle \Phi \rangle$$

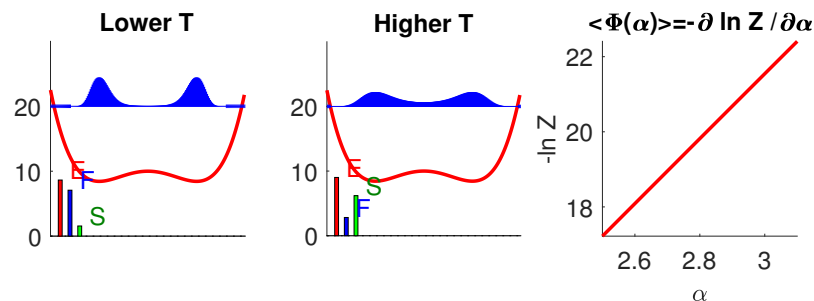
so that the energy can be seen to divide into two parts:

$$\langle \Phi \rangle = \underbrace{-\frac{1}{\alpha} \ln Z}_{F(p)} + \frac{1}{\alpha} S_{max}(p)$$

The second term is the energy tied up in heat, which is *not* available for useful work. The first part specifies the remaining energy that *is* available for useful work. This part is called "free energy"  $F(p)$ .

## Division and temperature

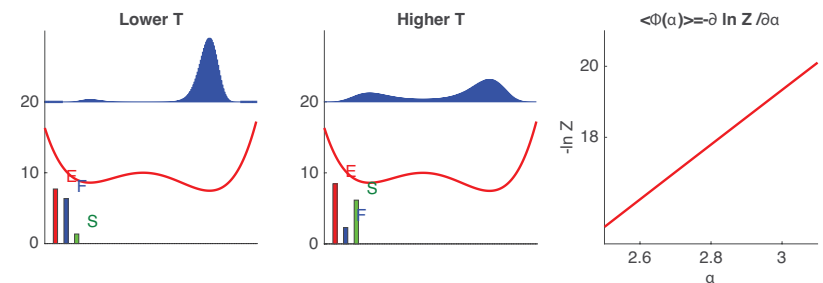
How the energy is divided – into free energy and entropy – depends on inverse temperature  $\alpha$  and the potential.



On the left, there is more free energy and less entropy. On the right, the situation is reversed.

## Division with another potential

Given another potential, we obtain a different division into free energy and entropy, which also depends on inverse temperature  $\alpha$ .



On the left, there is more free energy and less entropy. On the right, the situation is reversed.



## Minimizing energy against maximizing entropy

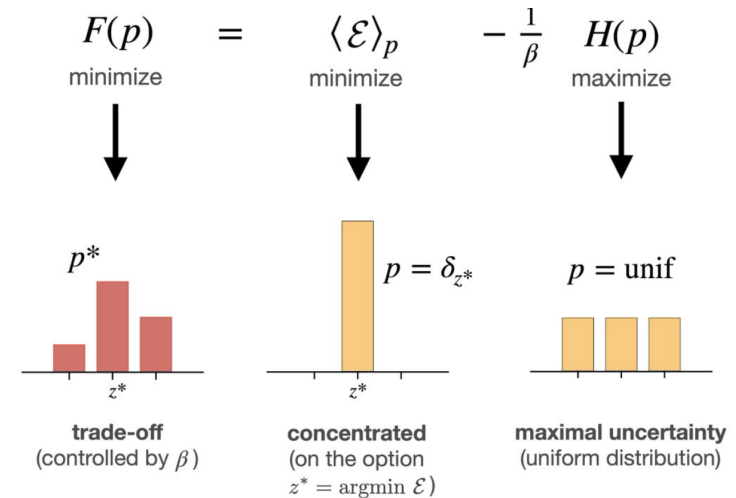
Another perspective is to consider the trade-off between energy  $\langle \Phi \rangle$  and entropy  $S$ . If the distribution  $p(x)$  was concentrated on the microstate of minimal energy,  $p(x) = \delta(x - x_{min})$ , then this would minimize  $\langle \Phi \rangle$ .

On the other hand, if the distribution was uniform, this would maximize uncertainty and entropy  $S(p)$ .

$$F(p) = \langle \Phi \rangle - \frac{1}{\alpha} S_{max}(p)$$

Importantly, minimizing the free energy  $F(p)$  identifies the equilibrium state that balances minimizing energy (but also lowering entropy) and maximizing entropy (but also raising energy)

Once again the trade-off is controlled by inverse temperature  $\alpha$ :



## Free energy difference (Gottwald & Braun)

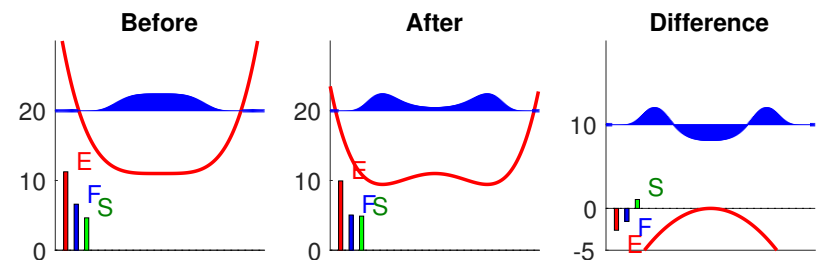
Starting from an initial steady-state distribution  $p_0(x)$  with potential  $\Phi_0(x)$ , energy  $\langle \Phi_0 \rangle$ , entropy  $S(p_0)$  and free energy  $F(p_0)$  we change the potential to  $\Phi(x) = \Phi_0(x) + \Delta\Phi(x)$  so that the new situation is now characterized by  $p(x)$ ,  $\Phi(x)$ ,  $S(p)$  and  $F(p)$ . The associated change in free energy  $\Delta F[p] = F[p] - F[p_0]$  is

$$\begin{aligned} \Delta F[p] &= F[p] - F[p_0] = \langle \Phi \rangle - \frac{1}{\alpha} S_{max}(p) - F[p_0] = \\ &= \langle \Delta\Phi \rangle_p + \frac{1}{\alpha} D_{KL}[p || p_0] \end{aligned}$$

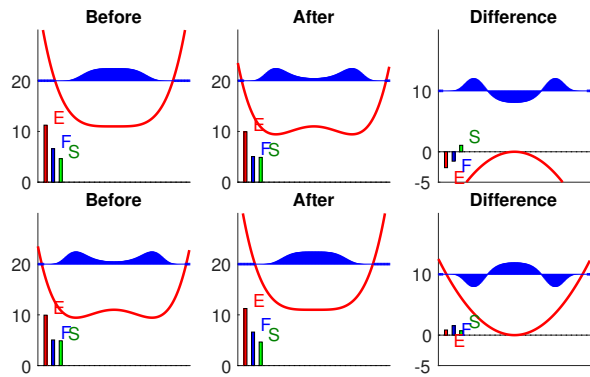
Note the sign change! Now it is the free energy difference that divides into energy difference *plus* differential entropy. Thus, to minimizing free energy we need to minimize both the energy difference and differential entropy.

## Free energy difference, ctd.

So what compromise does nature choose? Is it energy that is minimized? Is it entropy that is maximized?



## Free energy difference, ctd

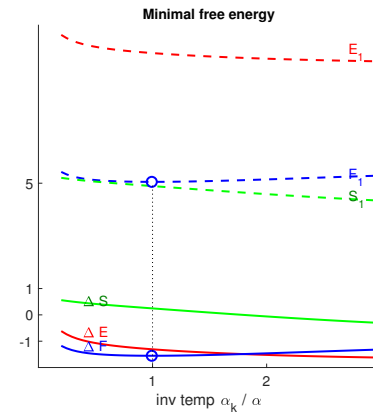


The KL divergence increases in both directions and thus is asymmetric. The energy difference is also asymmetric, because it has to compensate the asymmetry of the KL divergence.

## Free energy difference, conclusion

The associated change in free energy  $\Delta F[p] = F[p] - F[p_0]$  is

$$F[p] - F[p_0] = \langle \Delta \Phi \rangle_p - \langle \Phi_0 \rangle_{p_0} - \frac{1}{\alpha} [S_{\max}(p) - S_{\max}(p_0)]$$



If we compare different distributions  $p_k(x) = \frac{1}{Z_k} e^{-\alpha_k \Phi_1(x)}$  parametrized by hypothetical temperatures  $\alpha_k$ , we find that is is the distribution for  $\alpha_k = \alpha$  that minimizes the both the free energy of the final state  $F_1$  and the difference in free energy,  $\Delta F = F_1 - F_0$ , between initial and final states

## Conclusion

In summary, the distribution that maximizes the entropy under the constraint of a given energy and a given temperature, is necessarily also the distribution that minimizes the difference between energy and entropy (in units of energy). As this difference is the energy that can be extracted as useful work, we can call it free energy.

$$F(p) = \underbrace{-\frac{1}{\alpha} \ln Z}_{\text{minimize}} = \langle \Phi \rangle_p - \underbrace{\frac{1}{\alpha} S(p)}_{\text{maximize}}$$

Minimizing free energy is equivalent to maximizing entropy.

## 5. Derivation of Boltzmann distribution

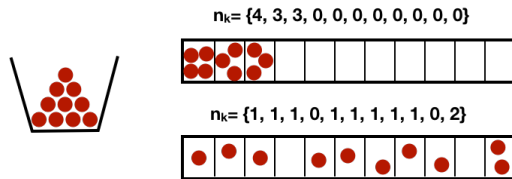


Edwin Jaynes, 1957, Maximum information principle: there is an unambiguous criterion for the "amount of uncertainty" represented by a probability distribution. This criterion agrees with our intuition that a broad distribution represents more uncertainty than a sharp one.

Physical systems are overwhelmingly likely to be encountered in a state with the broadest possible distribution.

## N particles in K compartments

Consider  $N$  particles distributed over  $K$  equal compartments. A particular distribution is defined by numbers  $n_k$  or probabilities  $p_k = \frac{n_k}{N}$  for each compartment, with  $k \in \{1, \dots, K\}$  and  $\sum_k p_k = 1$



The information theoretical entropy  $H = -\sum_k p_k \ln p_k$  is the number of ways in which a particular set of  $p_k$  may be realized (Proof: binomial statistics in the limit of large  $N$ , e.g., page 193, Jaynes, 1967). In the illustration above, the lower distribution has more entropy.

## Maximize entropy with Lagrangian multipliers

To solve for  $p_k$  with highest entropy, we maximize a linear combination of both conditions:

$$\frac{\partial}{\partial p_k} \left[ -\sum_k p_k \ln p_k - \lambda \sum_k p_k \right] = 0$$

where  $\lambda$  is an initially unknown constant:

$$\Rightarrow -\ln p_k - 1 - \lambda = 0 \quad \Leftrightarrow \quad p_k = e^{-\lambda-1}$$

It follows that

$$\sum_k p_k = 1 \quad \Rightarrow \quad p_k = e^{-\lambda-1} = \frac{1}{K}$$

Thus, as expected, a uniform distribution has the largest entropy.

## Physical entropy

In physical systems, entropy is denoted by  $S$  and expressed in terms of Boltzmann's constant  $k_B$ :

$$S = -k_B \sum_i p_i \ln p_i = -\sum_i p_i \ln p_i$$

We put  $k_B = 1$  to simplify notation. We seek to estimate  $p_i$  from certain known physical constraints. The maximum entropy estimate is *unbiased* in the sense that it provides the most likely  $p_i$  values for any given set of physical facts.

## Multiple constraints

In addition to  $\sum_i p_i = 1$ , we consider (possibly several) additional additive physical constraints  $\sum_i p_i f_i^{(k)} = f_k$  such as total mass, total energy, etc:

$$M_{tot} = \sum_i p_i M_i, \quad E_{tot} = \sum_i p_i E_i, \quad \dots$$

To identify the set of  $p_i$  that maximizes entropy while also satisfying all constraints, we maximize a linear combination of all terms

$$\frac{\partial}{\partial p_i} \left[ -\sum_i p_i \ln p_i - (\lambda - 1) \sum_i p_i - \lambda_e \sum_i p_i e_i \right] = 0$$

where  $\lambda$  and  $\lambda_e$  are initially unknown constants (Lagrangian multipliers).

## Partition function

After differentiation, we obtain

$$-\ln p_i - 1 - (\lambda - 1) - \lambda e_i = 0, \quad p_i = e^{-\lambda} e^{-\lambda e_i}$$

From  $\sum_i p_i = 1$  we obtain

$$e^{-\lambda} \underbrace{\sum_i e^{-\lambda e_i}}_Z = e^{-\lambda} Z = 1, \quad Z = e^\lambda, \quad \lambda = \ln Z$$

where

$$Z = \sum_i e^{-\lambda e_i}$$

is the *partition function* of the system, which depends on known microscopic constraints  $f_i^{(k)}$ , such as  $f_i^{(1)} = e_i$ ,  $f_i^{(2)} = m_i$ , ..., and on (as yet unknown) constants  $\lambda_e$ ,  $\lambda_m$ , ...

## Maximum entropy

When  $p_i$  satisfies  $p_i = e^{-\lambda} e^{-\sum_k \lambda_k f_i^{(k)}}$ , the entropy is maximal (proof on page 188, Jaynes, 1967). We can evaluate this entropy as

$$\begin{aligned} S_{max} &= -\sum_i p_i \ln p_i = \\ &= -\sum_i p_i \left[ -\lambda - \sum_k \lambda_k f_i^{(k)} \right] \\ &= \lambda \sum_i p_i + \sum_k \lambda_k \sum_i p_i f_i^{(k)} \\ &= \lambda + \sum_k \lambda_k f_k = \ln Z + \sum_k \lambda_k f_k \end{aligned}$$

This expression becomes meaningful and interesting when our physical constraints are made concrete.

## Other Lagrangian multipliers (optional)

To determine the other unknown constants  $\lambda_k$ , we use constraints  $\langle f_i^{(k)} \rangle = f_k$  to obtain

$$\begin{aligned} f_k &= \sum_i p_i f_i^{(k)} = e^{-\lambda} \sum_i e^{-\sum_l \lambda_l f_i^{(l)}} \cdot f_i^{(k)} = \\ &= e^{-\lambda} \left( -\frac{\partial}{\partial \lambda_k} \right) \left[ \sum_i e^{-\sum_l \lambda_l f_i^{(l)}} \right] \\ &= \frac{1}{Z} \left( -\frac{\partial}{\partial \lambda_k} \right) Z = -\frac{\partial \ln Z}{\partial \lambda_k} \end{aligned}$$

Thus, all Lagrangian multipliers can be obtained by matching derivatives of partition function  $Z$  to known macroscopic constraints  $\langle f_i^{(k)} \rangle = f_k$ , such as  $\langle M_i \rangle = f_1 = M_{tot}$ ,  $\langle E_i \rangle = f_2 = E_{tot}$ , ...

## Entropy and macroscopic constraints

Considering the dependence of entropy on macroscopic constraints  $\langle f_i^{(k)} \rangle = f_k$ , yields a physical interpretation of the Lagrangian multipliers:

$$S_{max} = \ln Z + \sum_k \lambda_k f_k, \quad Z = \sum_i e^{-\sum_k \lambda_k f_i^{(k)}}$$

$$\frac{\partial S_{max}}{\partial f_k} = \lambda_k$$

The constants  $\lambda_k$  capture the proportionality between entropy and macroscopic constraints such as average mass or average energy.

## Boltzmann distribution

When we constrain the total internal energy  $E_{tot}$  or  $U$  in Boltzmann's notation

$$f_i^{(1)} = E_i, \quad f_1 = \sum_i p_i E_i = \langle E_i \rangle = U, \quad \lambda_1 = \beta$$

we obtain the Boltzmann distribution and partition function

$$p_i = e^{-\lambda - \beta E_i} = \frac{1}{Z} e^{-\beta E_i}, \quad Z = \sum_i e^{-\beta E_i}$$

and the maximal entropy becomes

$$S_{max} = \ln Z + \lambda_1 f_1 = \ln Z + \beta U$$

## Free energy

Rearranging

$$S_{max} = \ln Z + \beta U$$

yields the free energy

$$F = -\frac{1}{\beta} \ln Z = U - \frac{1}{\beta} S_{max}$$

which is that part of internal energy  $U$  that is not tied up in entropy and is available to do useful work. Here  $1/\beta = T$  is absolute temperature.

## Compare generative models

Given observation  $\mathbf{u}$  and model parameters  $\theta$ , we typically wish to optimize  $\theta$  by maximizing our objective function, namely,  $p(\mathbf{u}|\theta)$ , the probability of generating  $\mathbf{u}$  from  $\theta$ .

If there are many alternative explanatory states  $\alpha$ , we really need to maximize the sum

$$p(\mathbf{u}|\theta) = \sum_{\alpha} p(\mathbf{u}|\alpha, \theta)$$

The situation becomes formally analogous to statistical physics, if we define a microscopic energy of explanation  $\alpha$  as

$$E_{\alpha}(\mathbf{u}, \theta) \equiv -\ln p(\mathbf{u}|\alpha, \theta)$$

with an associated probability of explanation (Boltzmann distribution)

$$P_{\alpha} \equiv \frac{e^{-E_{\alpha}}}{\sum_{\alpha} e^{-E_{\alpha}}}$$

With these definitions, the likelihood of observations is

$$p(\mathbf{u}|\theta) = \sum_{\alpha} e^{-E_{\alpha}}$$

and the maximum entropy is

$$-\sum_{\alpha} P_{\alpha} \ln P_{\alpha} = \ln \sum_{\alpha} e^{-E_{\alpha}} + \sum_{\alpha} P_{\alpha} E_{\alpha}$$

so that the 'surprise' (negative log likelihood) of observations becomes

$$-\ln p(\mathbf{u}|\theta) = \underbrace{-\ln \left[ \sum_{\alpha} e^{-E_{\alpha}} \right]}_{\text{free energy}} = \underbrace{\sum_{\alpha} P_{\alpha} E_{\alpha}}_{\text{energy}} - \underbrace{\left( -\sum_{\alpha} P_{\alpha} \ln P_{\alpha} \right)}_{\text{entropy}}$$

## Overall summary

- ▶ Brains try to predict the environment with ‘causal models’, in which latent causes closely approximate true causes.
- ▶ The best test for ‘causal models’ is by interacting with the environment.
- ▶ Variational ‘free energy’ has two parts: divergence of recognition and surprise. Causal models can be fitted to observations by minimizing both parts (with EM).
- ▶ Friston speculates that both minimizations could be performed by a dynamic of brain activity.
- ▶ A physical change in ‘free energy’ has two parts: divergence of distribution and change in energy. Nature maximizes entropy or, equivalently, minimizes free energy.

## Next: Principal component analysis